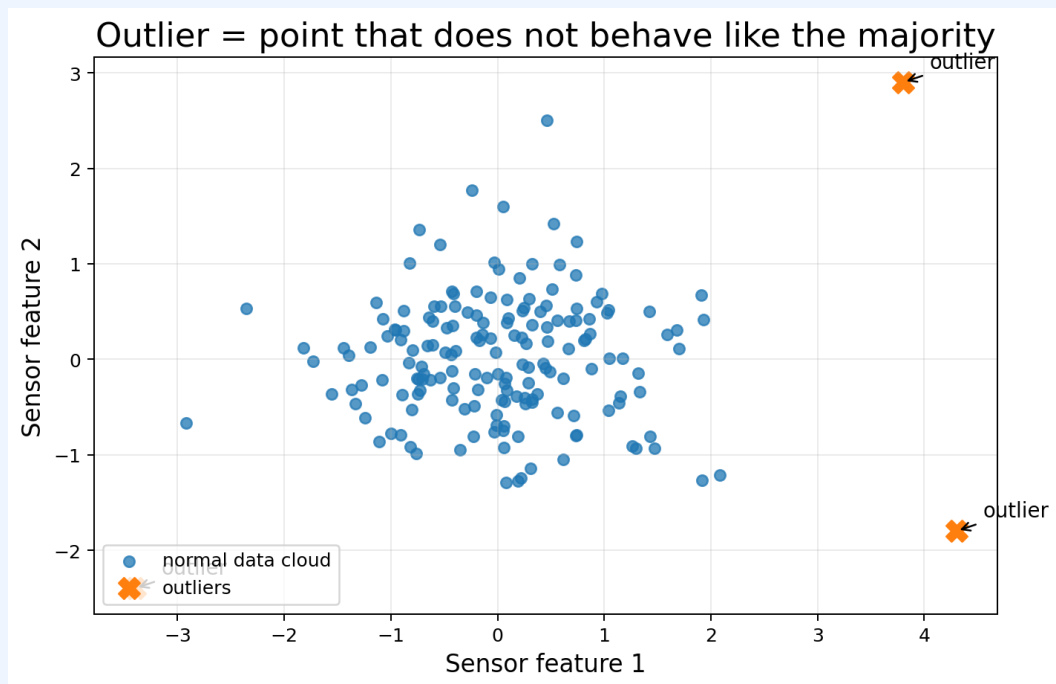


Outlier Detection

Zero-to-Hero Professional Visual Theory Notes

A simplified, detailed and visual-first guide covering Z-Score, IQR, Boxplots, LOF, LoOP, ABOD, Isolation Forest, CBLOF, PCA, Jaccard, Autoencoders, reconstruction error and evaluation metrics.



Outlier detection starts with one simple question: which points do not behave like the majority?

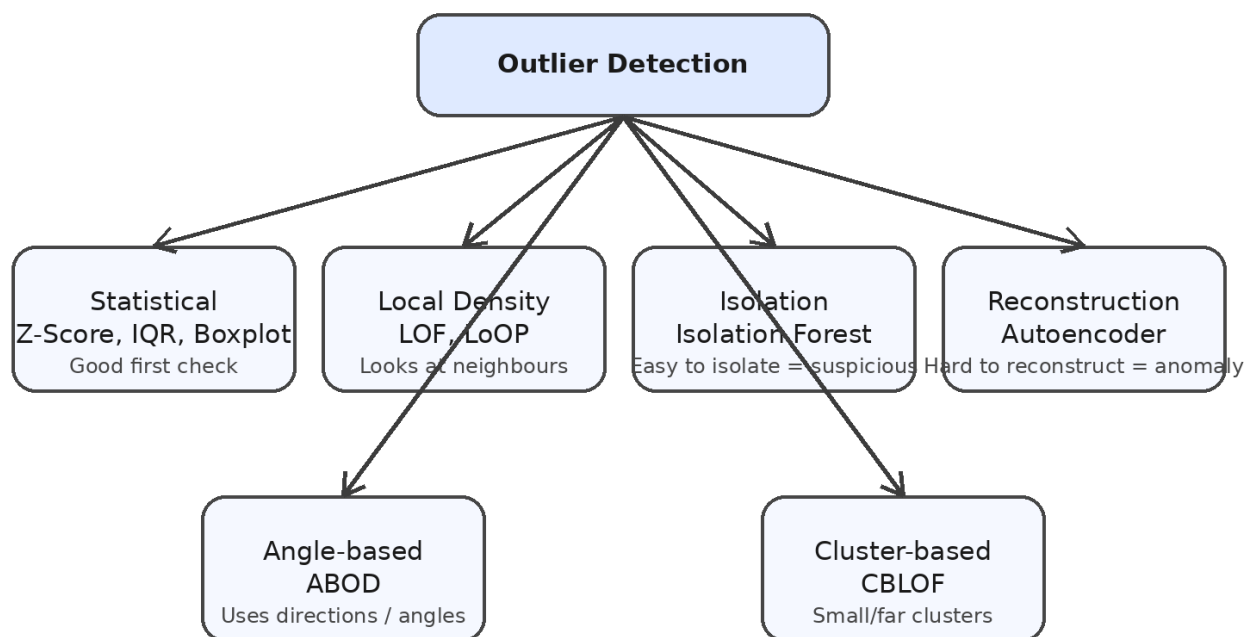
Prepared from your uploaded educx Day 4 task sheet, theory PDFs, lecture notes and supplied visual intuition images.

How to use these notes

Goal: understand the complete theory visually before writing code. Each method is explained with one simple idea, one visual intuition, and one practical interpretation.

Full roadmap

Outlier Detection Method Map



The complete method map: beginner statistical methods, intermediate algorithmic methods, extra theory methods, and expert reconstruction methods.

Contents

#	Topic	Main learning
1	What is an outlier?	Foundation and practical examples
2	Outlier vs anomaly	When to keep, investigate, or remove
3	Why outliers affect statistics	Mean, median, range, IQR
4	Z-Score	Statistical distance from the mean
5	IQR and Boxplot	Robust visual detection
6	Why advanced algorithms are needed	Multivariate and high-dimensional issues
7	LOF	Local density outlier detection
8	LoOP	Local outlier probability
9	ABOD	Angle-based outlier detection

#	Topic	Main learning
10	Isolation Forest	Random isolation and path length
11	PCA and Jaccard	Visualization and method agreement
12	CBLOF	Cluster-based scoring
13	Autoencoder	Reconstruction-based anomaly detection
14	Metrics and deployment	Precision, Recall, F1, real-time comparison

1. What is an outlier?

An **outlier** is a data point that is clearly different from the normal pattern of the dataset. In a picture, most points form a normal cloud, while outliers sit far away or behave differently.

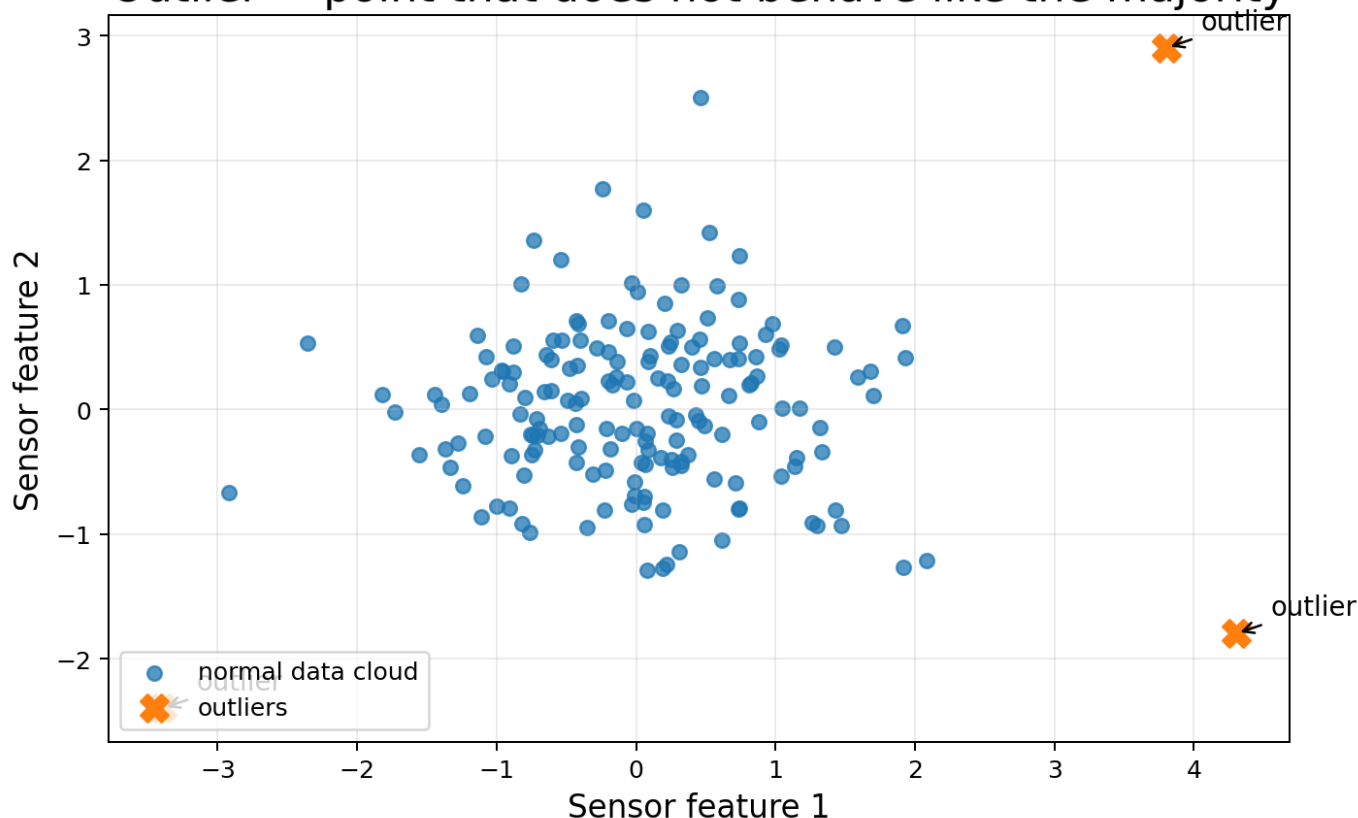
Simple meaning

Most rows follow a similar pattern. An outlier is a row that breaks this pattern.

Business meaning

It may be a data error, fraud attempt, machine failure signal, rare customer behavior, or new important event.

Outlier = point that does not behave like the majority



Visual example: the crossed points are unusual compared with the main data cloud.

Memory trick: Outlier = “This point does not behave like the majority.”

Practical examples

Area	Normal pattern	Outlier
Machine sensors	Temperature usually 70-80 C	One reading is 170 C
Fraud	Customer usually spends 20-80 EUR	One transaction is 5,000 EUR
Marketing	CTR usually 2-4%	One campaign has 35% CTR
Inventory	Daily demand usually 10-30 units	One day demand is 800 units
Network security	Normal traffic pattern	Attack-like traffic burst

Important: an outlier is not automatically bad data. It can be an error, but it can also be the most important signal in the dataset.

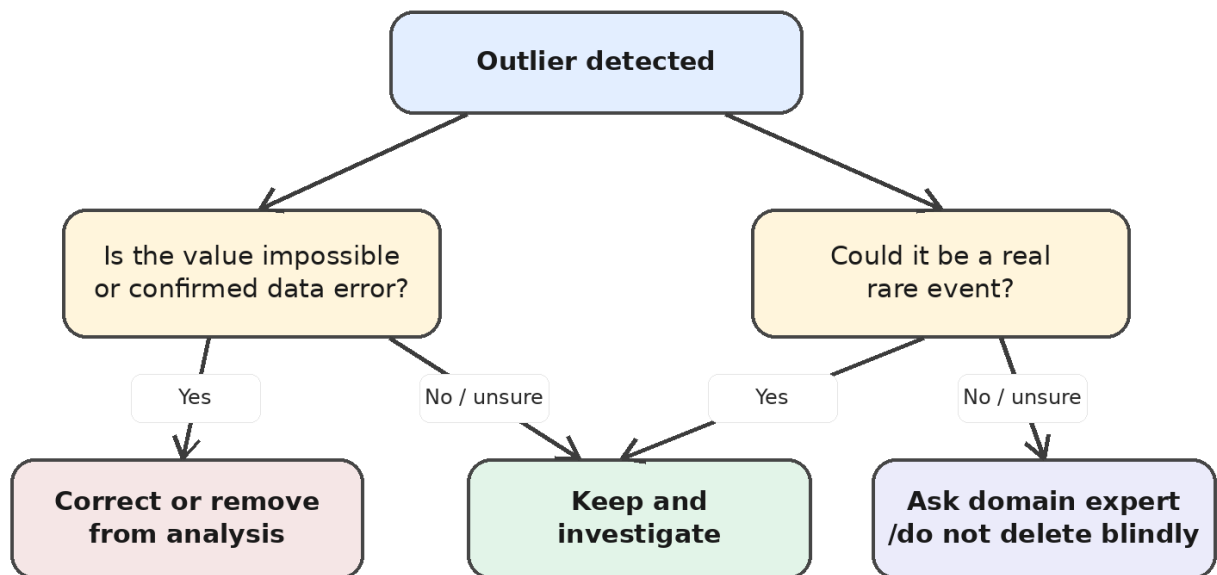
2. Outlier vs anomaly, and when to remove

Term	Simple meaning	Example
Outlier	Looks unusual in the data	A sensor value is far away from the usual range
Anomaly	Unusual and important or suspicious	The unusual sensor value indicates real machine failure

The same value can have different meanings depending on context. A value of 0 in a sensor column may mean a broken sensor, a switched-off machine, a missing value, or a valid process state.

Core rule: detect first, investigate second, and remove only when justified by domain knowledge.

Keep or remove an outlier?



Decision flow: never delete an outlier only because it looks strange.

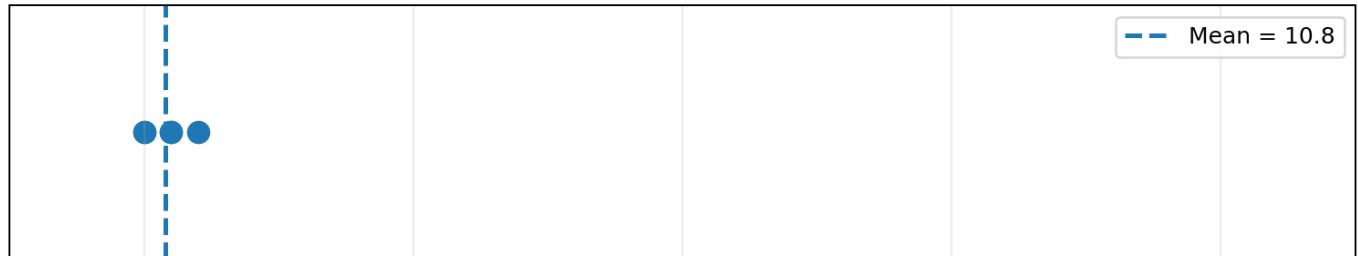
Keep vs remove decision table

Situation	Best action	Reason
Impossible value	Correct or remove	Likely data quality issue
Confirmed broken sensor or ETL error	Correct or remove	Not real business behavior
Fraud transaction	Keep and label/investigate	This is the signal we want to catch
Machine failure signal	Keep and analyze	Important rare event
Unknown reason	Investigate with domain expert	Do not guess

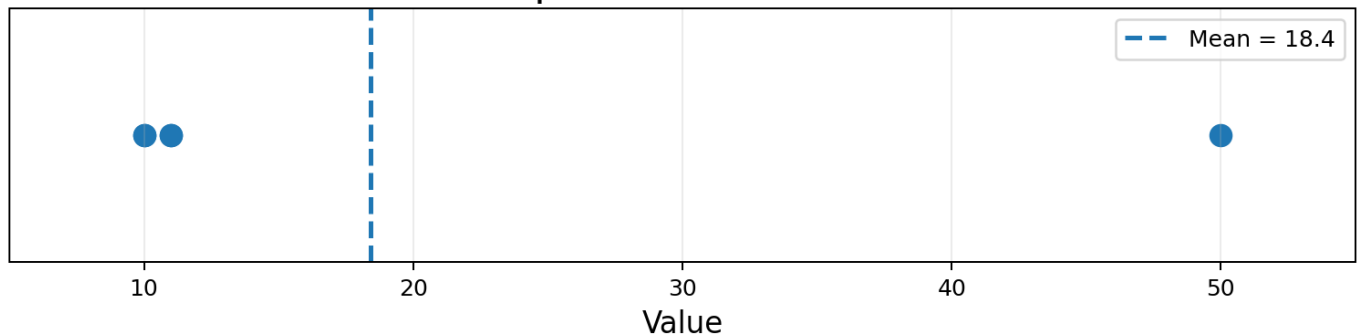
3. Why outliers affect statistics

Outliers can strongly distort summary statistics. The mean is especially sensitive because every value contributes to the average.

Before: no extreme value



After: one extreme value pulls the mean



One extreme value pulls the mean toward itself. This is why the mean is sensitive to outliers.

Mean

Average value. Easy to understand, but sensitive to extreme values.

Median

Middle value after sorting. More robust when outliers exist.

Range

Maximum minus minimum. Very sensitive to outliers.

IQR

Spread of the middle 50%. Much more robust to outliers.

Memory trick: Mean can be pulled. Median and IQR are more stable.

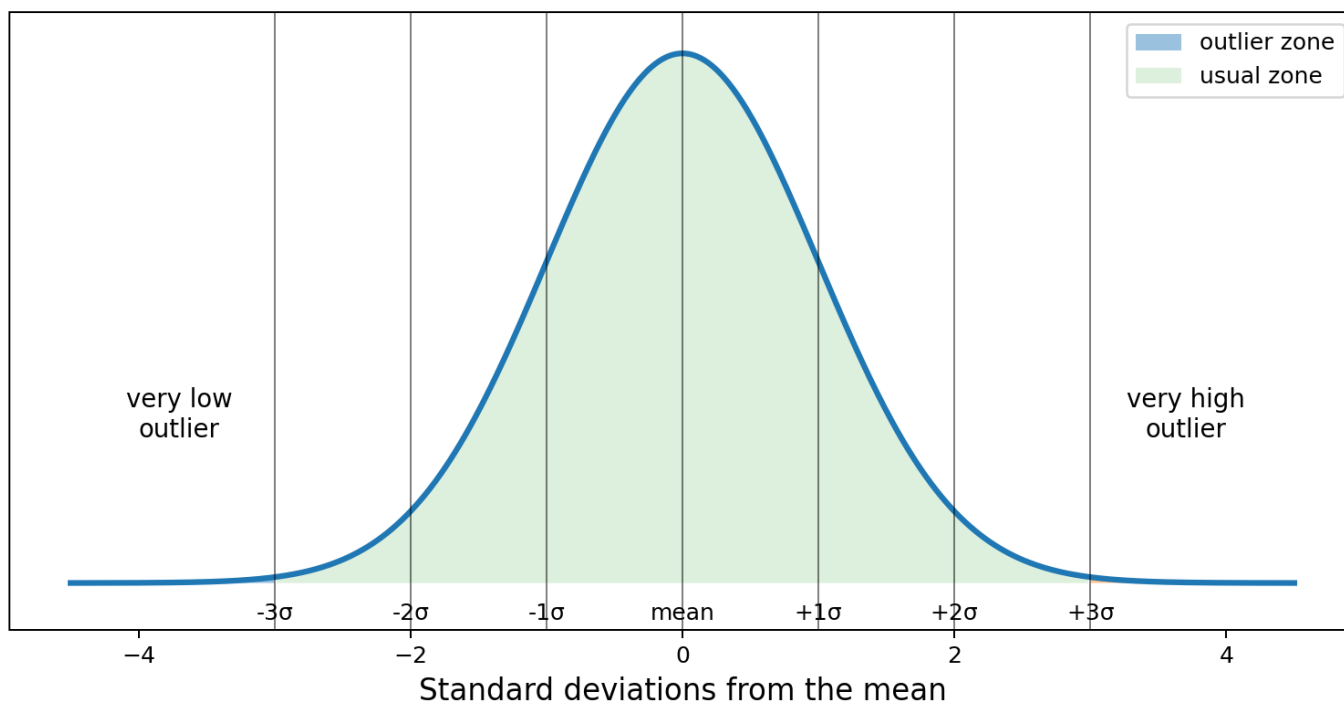
This is why outlier handling can change reports, dashboards, KPI averages, model training, and business decisions.

4. Z-Score - statistical distance from normal

A **Z-Score** tells you how far a value is from the mean, measured in standard-deviation units.

$$z = (x - \text{mean}) / \text{standard deviation}$$

Z-Score: how far a value is from the mean



Z-Score visual: values beyond about -3 or +3 standard deviations are usually treated as statistical outliers.

Simple example: machine temperature

Temperature	Mean	Std	Z-Score	Interpretation
75	75	5	0	Exactly normal
80	75	5	1	Slightly high
90	75	5	3	Very high
100	75	5	5	Strong outlier

Day 4 rule: mark a row as an outlier if at least one numeric feature has absolute Z-Score greater than 3.

Strengths and weaknesses

Strengths	Weaknesses
Simple, fast, easy to explain, good first screening method	Works best for roughly normal data, sensitive to extreme values, checks columns separately, and may miss multivariate anomalies

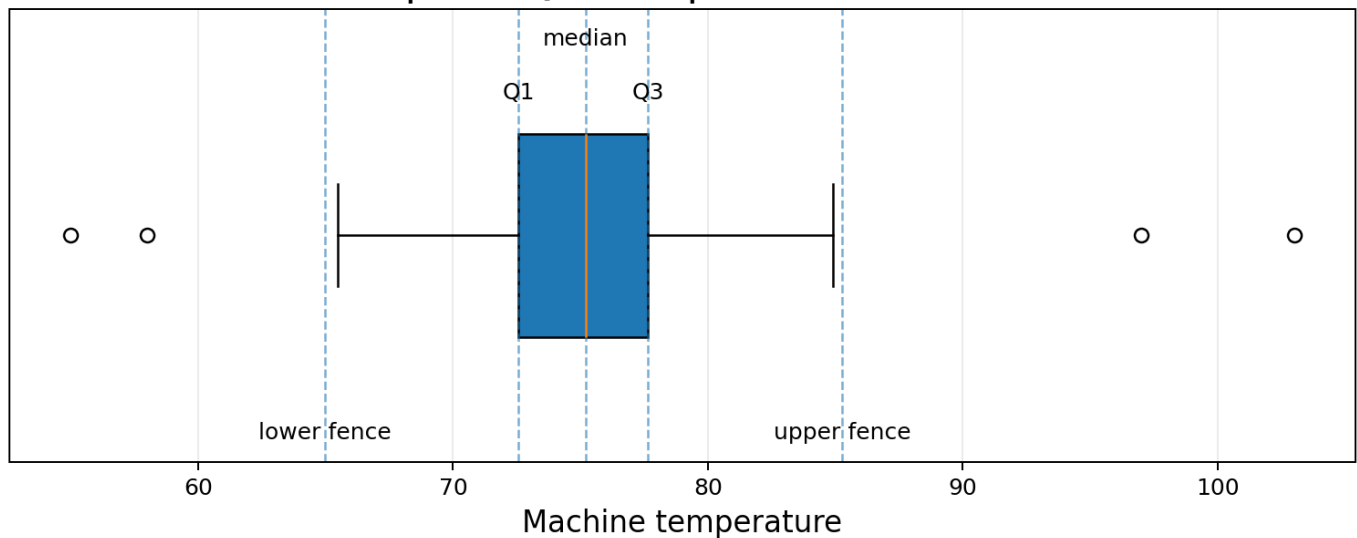
5. IQR and Boxplot - robust visual detection

The **IQR** is the spread of the middle 50% of the data. It uses quartiles instead of mean and standard deviation, so it is more robust to extreme values.

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower fence} = Q1 - 1.5 \times \text{IQR} \mid \text{Upper fence} = Q3 + 1.5 \times \text{IQR}$$

Boxplot: IQR and potential outliers



Boxplot visual: the box is the middle 50%; points outside the fences are potential outliers.

Memory trick: Z-Score uses mean and std. Boxplot uses quartiles and IQR.

Method	Best when	Potential problem
Z-Score	Data is roughly bell-shaped	Mean and std are sensitive to outliers
IQR / Boxplot	You want a robust visual method	May still flag valid extreme business cases

6. Why advanced algorithms are needed

Statistical methods are useful, but they often look at one feature at a time. Real anomalies can be hidden in the relationship between many features.

Single-column outlier

Temperature alone is too high. Z-Score or boxplot can catch it easily.

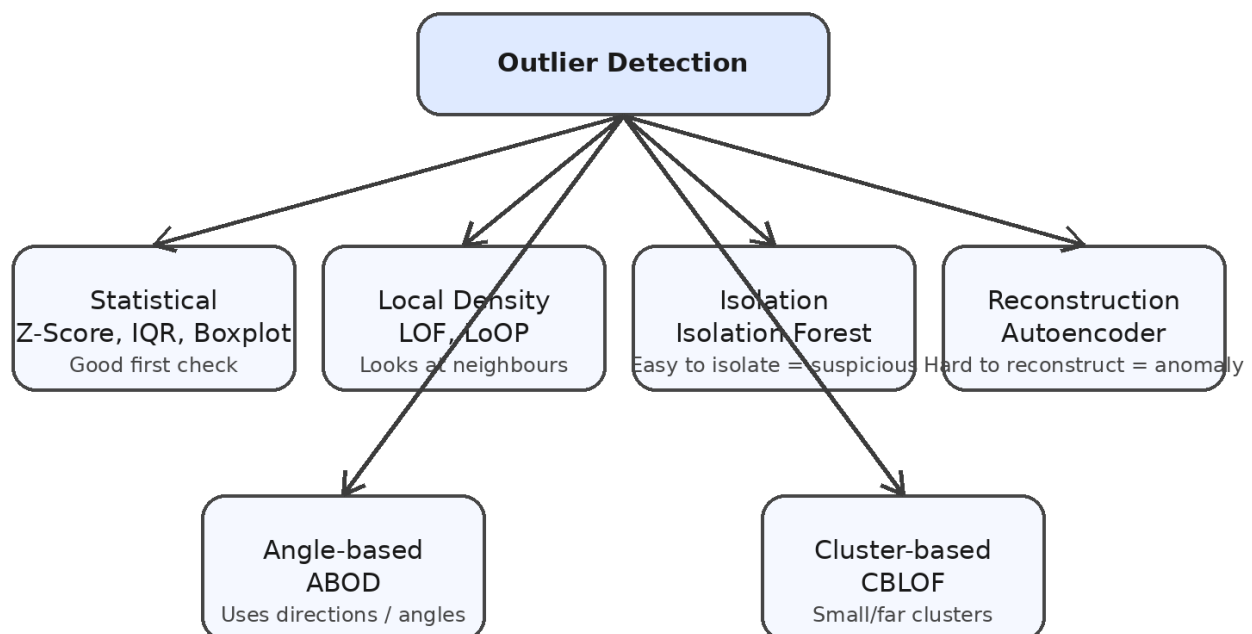
Multivariate outlier

Temperature, pressure and vibration individually look normal, but their combination is strange.

High-dimensional problem: as the number of features increases, distance-based reasoning can become weaker because many points start looking similarly far away.

This is why Day 4 moves from Z-Score to algorithmic methods such as Isolation Forest and LOF. It is also why ABOD focuses on angles instead of relying only on distances.

Outlier Detection Method Map



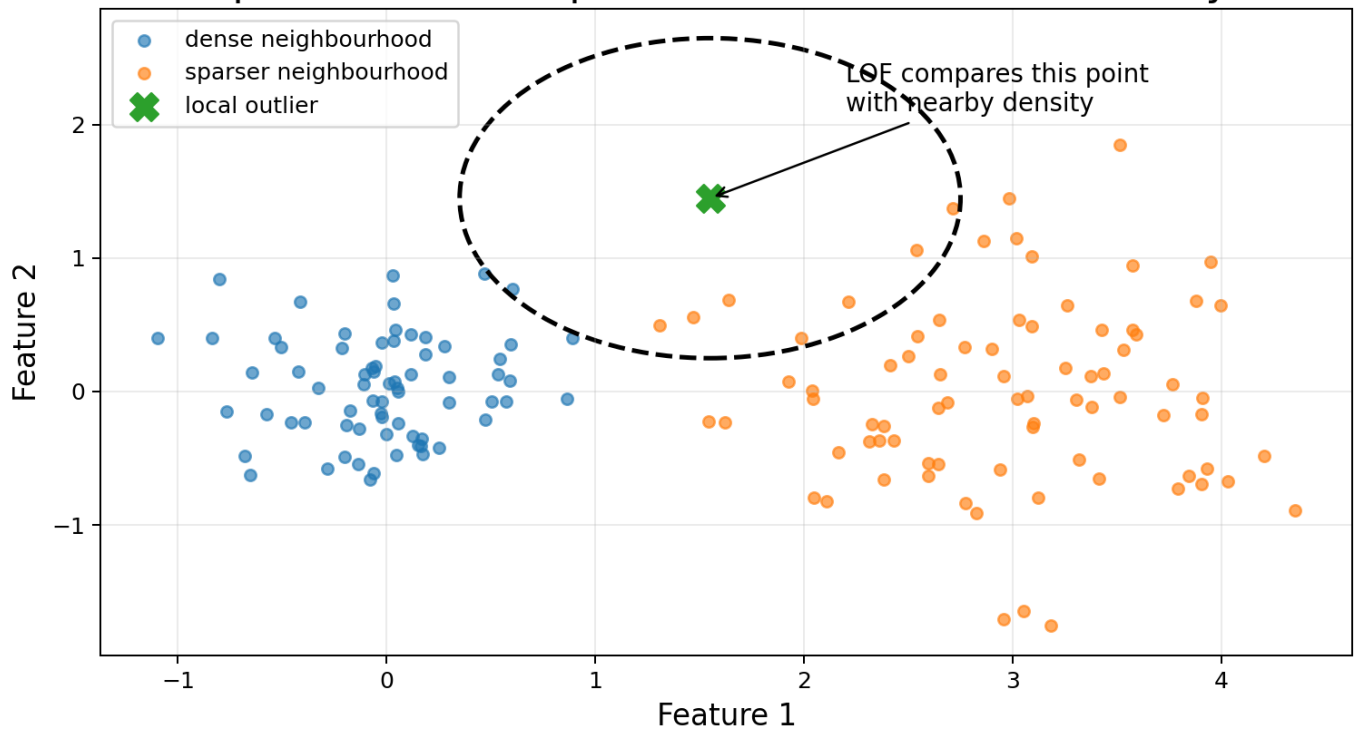
Advanced methods detect different definitions of unusual behavior.

7. LOF - Local Outlier Factor

LOF asks whether a point is in a region where its local density is much lower than the density of its neighbours.

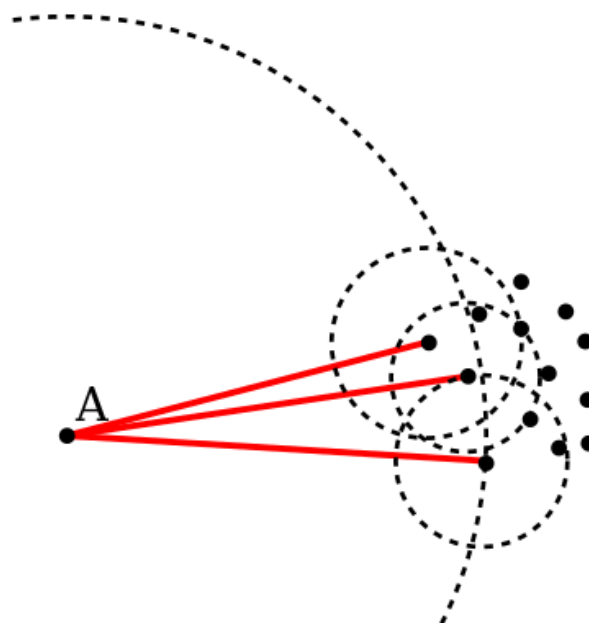
Memory trick: LOF = “Am I less dense than my neighbours?”

LOF: a point can be suspicious because its local density is low



Simplified LOF diagram: the suspicious point is judged relative to nearby points, not the whole dataset.

Actual uploaded intuition image



Uploaded LOF intuition image: local density is the key idea. Points in sparse local regions can be suspicious.

How LOF works

Step	Simple explanation	
1	Choose k nearest neighbours for each point	
2	Estimate the local density around each point	
3	Compare each point density with neighbour densities	
4	If the point is much less dense, it becomes suspicious	
Strengths		Weaknesses
Good for local anomalies and regions with different densities		Sensitive to neighbour parameter k and harder to explain than Z-Score

8. LoOP - Local Outlier Probabilities

LoOP is related to local density methods, but it tries to make scores easier to interpret by producing a probability-like value between 0 and 1.

0 = likely normal | 1 = likely outlier

Why LoOP exists

Many methods output scores that are hard to interpret. For example, what does LOF = 2.7 mean for a business user?

What LoOP improves

It normalizes local outlierness into a probability-like range, making threshold communication easier.

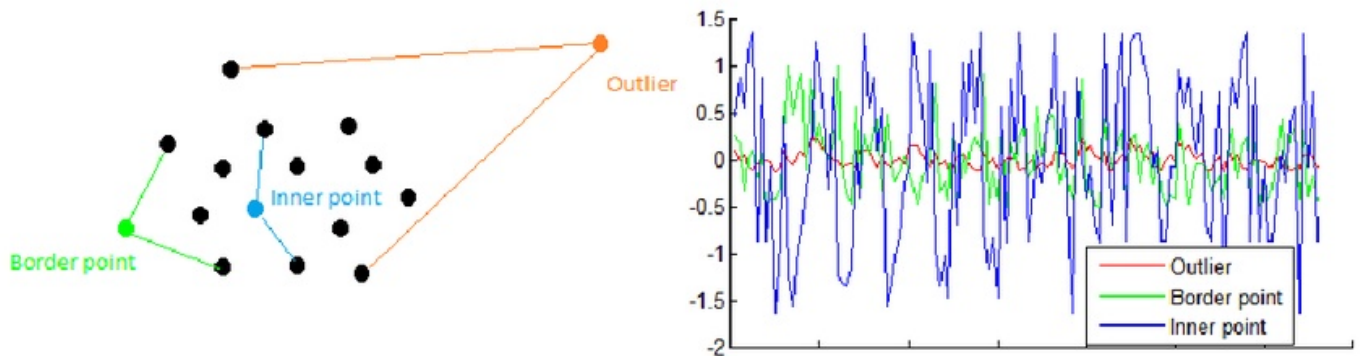
Simple explanation: LOF says “more or less suspicious”. LoOP tries to say “how likely suspicious” on a 0-to-1 scale.

Score	Interpretation
Near 0	Likely normal
Around 0.5	Uncertain or moderately suspicious
Near 1	Likely outlier

9. ABOD - Angle-Based Outlier Detection

ABOD looks at angles between vectors from one point to other points. It is especially useful as theory for high-dimensional data.

Memory trick: ABOD = “Look at directions, not only distances.”



Actual uploaded ABOD intuition image: inner points, border points, and outliers have different angle patterns.

Point inside a cluster

Other points surround it from many directions. The angles vary a lot. This usually looks normal.

Point outside a cluster

Most other points lie in one general direction. Angle variation is smaller. This looks suspicious.

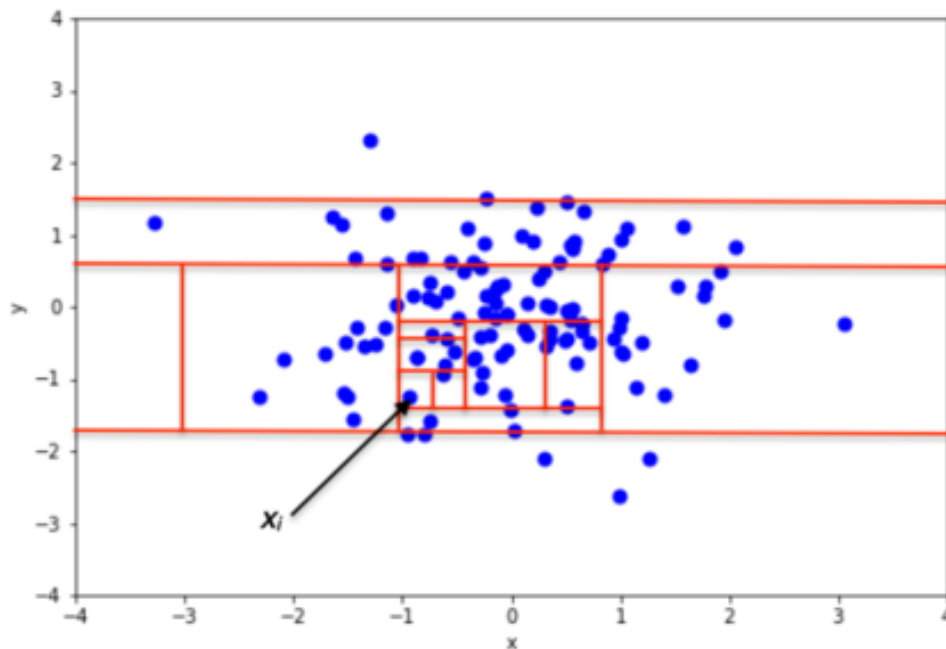
Why this matters: in high-dimensional data, raw distances can become less meaningful. ABOD reduces this problem by using angle variance.

Strengths	Weaknesses
Strong theoretical idea for high-dimensional outlier detection	Can be computationally heavier and less common in basic sklearn workflows

10. Isolation Forest

Isolation Forest detects anomalies by asking: how quickly can a point be isolated using random splits?

Memory trick: anomaly = few and different = easy to isolate.



Actual uploaded Isolation Forest intuition image: unusual points can often be separated in fewer random splits.

Simple process

Step	Explanation
1	Randomly choose a feature
2	Randomly choose a split value
3	Split the data into two parts
4	Repeat until points are isolated
5	Build many trees and average the path length

Normal point

Usually buried inside a dense group, so it needs many random cuts to isolate.

Outlier

Often separated early because it is far, rare, or different. Short path = suspicious.

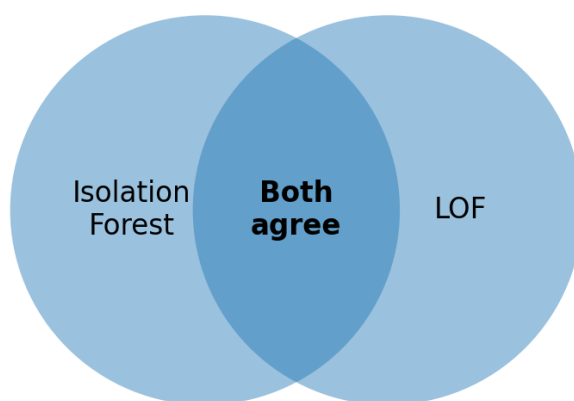
Day 4 parameter: contamination = 0.05 means the model is told to expect about 5% of the rows to be outliers.

11. LOF vs Isolation Forest

Question	LOF	Isolation Forest
Main question	Is my local density lower than my neighbours?	Can I be isolated quickly?
Best at	Local anomalies	General anomaly screening at scale
Uses distance/density?	Yes	Not directly
Main parameter	Number of neighbours	Contamination and number of trees
Business explanation	This row is strange compared with nearby similar rows.	This row separates from the dataset very quickly.

Important: if LOF and Isolation Forest disagree, it does not mean one is wrong. They define “unusual” differently.

Jaccard coefficient: method agreement



Jaccard = intersection / union

Jaccard helps quantify how much two methods agree on detected outliers.

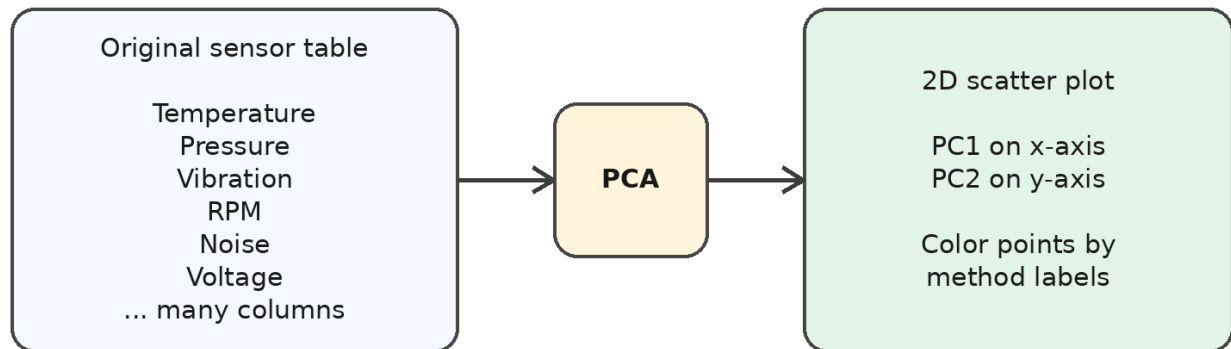
$$\text{Jaccard}(A, B) = \text{intersection}(A, B) / \text{union}(A, B)$$

Jaccard value	Meaning
0	No overlap
0.3	Some overlap, methods differ a lot
0.7	Strong overlap
1	Perfect agreement

12. PCA for outlier visualization

PCA reduces many sensor columns into two components so we can draw a 2D scatter plot.

PCA Visualization: many sensor columns -> 2D plot



Purpose: visualization, not magic. It helps us see where different methods flag outlier

PCA turns many sensor features into PC1 and PC2 for visualization of algorithm results.

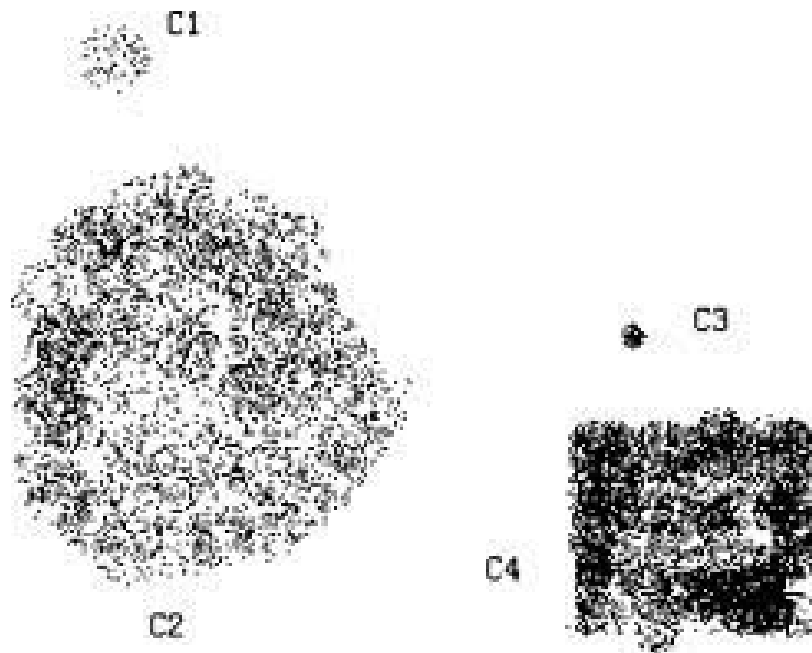
Important: PCA in this exercise is mainly for visualization. It helps us see where methods flag points.

What to look for in the PCA plot

- points both methods agree on
- points only LOF flags
- points only Isolation Forest flags
- whether anomalies sit far away or on cluster borders
- whether the result makes business sense

13. CBLOF - Cluster-Based Local Outlier Factor

CBLOF uses clustering to identify points that belong to small clusters or are far from important cluster centers.



Actual uploaded CBLOF intuition image: outlier score is related to cluster membership and distance to cluster centers.

Simple process

Step	Explanation
1	Cluster the dataset
2	Separate large clusters from small clusters
3	Calculate distance to cluster centers
4	Small/far cluster patterns get higher outlier scores

Memory trick: CBLOF = suspicious cluster membership + distance from cluster centers.

Good for	Weakness
Data that naturally forms groups or clusters	Depends on clustering quality and cluster assumptions

14. Autoencoder anomaly detection

An **autoencoder** is a neural network that learns to reconstruct its own input. It has an encoder, a bottleneck, and a decoder.

Autoencoder Anomaly Detection



Train on normal data only. Normal points reconstruct well. Anomalies reconstruct badly -> high rec

Architecture from the Day 4 expert task: 64 -> 32 -> 16 -> 32 -> 64.

Normal point

The model has learned this pattern, so reconstruction is good. Error is low.

Anomalous point

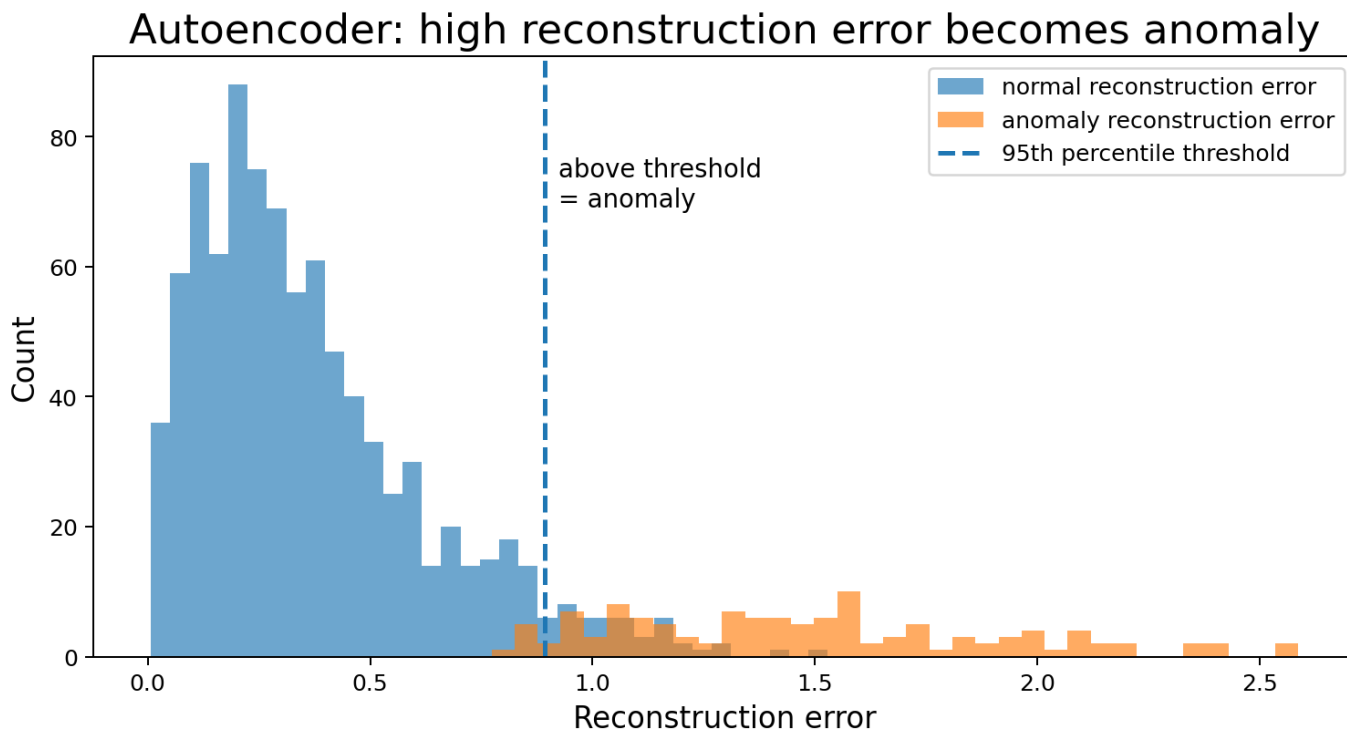
The model has not learned this pattern well, so reconstruction is poor. Error is high.

Reconstruction error = difference between original input and reconstructed output

Why train only on normal data? Because the goal is to learn normal behavior. If anomalies are included, the model may learn to reconstruct them too.

15. Reconstruction error and threshold

After training, pass data through the autoencoder and calculate reconstruction error for each row. High error means the row does not look like the normal data the model learned.



Rows above the 95th percentile threshold are classified as anomalies.

95th percentile threshold

The Day 4 expert task uses the 95th percentile. This means 95% of normal reconstruction errors are below this value. Rows above this value are treated as anomalies.

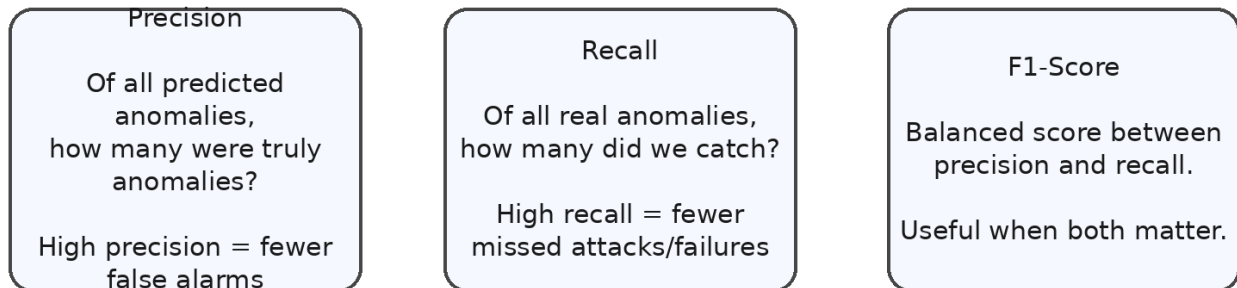
Memory trick: If the autoencoder cannot rebuild it well, it probably does not look normal.

Row	Reconstruction error	Threshold	Prediction
A	0.03	0.12	Normal
B	0.08	0.12	Normal
C	0.31	0.12	Anomaly

16. Precision, Recall and F1

When labels are available, we evaluate anomaly predictions using Precision, Recall, and F1.

Precision, Recall and F1 - simple meaning



Fraud / security often needs high recall. Business alert systems often also need good precision to

Simple business meaning of Precision, Recall, and F1.

Metric	Formula	Business meaning
Precision	$TP / (TP + FP)$	Of all alerts, how many were correct?
Recall	$TP / (TP + FN)$	Of all real anomalies, how many did we catch?
F1	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$	Balanced score between false alarms and missed anomalies

Real-world tradeoff: fraud and security often prefer high recall, but business alert systems also need good precision to avoid too many false alarms.

17. Autoencoder vs Isolation Forest

Aspect	Autoencoder	Isolation Forest
Core idea	Learn normal patterns and flag high reconstruction error	Randomly isolate points and flag short path length
Complexity	Higher	Lower
Best for	Complex nonlinear patterns	Fast baseline for tabular anomaly detection
Data need	Works best with normal training data	Can work as an unsupervised detector
Deployment	More engineering, monitoring, tuning	Simpler and faster to deploy
Explainability	Moderate: reconstruction error	Moderate: isolation/path length score

Use Isolation Forest when...

You need a quick baseline, simple deployment, tabular anomaly detection, and lower engineering complexity.

Use Autoencoder when...

Patterns are nonlinear, you have enough normal data, and you can tune and monitor a neural model.

18. One-page cheat sheet

Method	One-line explanation	Memory trick
Z-Score	How many standard deviations from mean?	Far from mean = suspicious
IQR / Boxplot	Uses quartiles and middle 50% spread	Outside whiskers = potential outlier
LOF	Compares local density with neighbours	Less dense than neighbours
LoOP	Local outlier score as probability-like value	0 normal, 1 outlier
ABOD	Uses angle variation between points	Directions, not only distances
Isolation Forest	Uses random splits to isolate anomalies	Easy to isolate = suspicious
CBLOF	Uses clusters, cluster sizes and centroid distances	Small/far cluster = suspicious
Autoencoder	Learns normal reconstruction	High reconstruction error = anomaly
PCA	Projects many features into 2D	Visual comparison tool
Jaccard	Measures overlap between outlier sets	Agreement score

Final practical principle: Outlier detection is not only math. It is math + visualization + domain knowledge + business decision.

19. Interview-style answers

What is an outlier?

An outlier is a data point that deviates strongly from the normal pattern. It may be an error, a rare event, or an important signal such as fraud or machine failure.

Why should we not delete all outliers?

Because outliers can represent important events. We should remove them only when we have domain evidence that they are invalid or caused by data quality issues.

What is the difference between Z-Score and LOF?

Z-Score checks how far a value is from the mean in one feature. LOF checks whether a point has lower local density than its neighbours, so it can detect local anomalies.

How does Isolation Forest detect anomalies?

It randomly splits data to isolate points. Anomalies are usually few and different, so they are often isolated with fewer splits and shorter path lengths.

How does an autoencoder detect anomalies?

It learns to reconstruct normal data. If a new point has high reconstruction error, it does not match learned normal patterns and is flagged as anomalous.

Why compare methods with Jaccard?

Because different algorithms define outliers differently. Jaccard tells us how much their detected outlier sets overlap.

20. Sources used from your uploaded material

- **educx Module 2 Machine Learning - Learning Day 4:** task structure for Z-Score, boxplot, Isolation Forest, LOF, PCA, Jaccard, autoencoder, reconstruction error and evaluation.
- **Angle-Based Outlier Detection in High-dimensional Data:** high-dimensional distance challenges and angle variance intuition.
- **Isolation Forest paper:** “few and different” anomaly intuition, short path length, random isolation trees and scalability.
- **LoOP - Local Outlier Probabilities:** motivation for interpretable local outlier scores in the range 0 to 1.
- **Tag 4 Outlier/Anomaly notes:** keep vs remove cautions, IQR robustness and practical handling of outliers.
- **Uploaded intuition images:** LOF, Isolation Forest, ABOD and CBLOF visuals included directly in these notes.

Next step after theory: begin Beginner coding with one clean cell at a time: imports, constants, dataset loading, and then Z-Score calculation.